

Prevalence and determinants of adherence to statin therapy: a systematic review and meta-analysis

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Abstract

Adherence to statin therapy remains suboptimal and is significantly influenced by demographic and clinical factors. Targeted strategies are needed to improve adherence, particularly in high-risk groups. Adherence to statin therapy is a key determinant of cardiovascular risk reduction, yet real-world adherence remains uncertain across populations. We conducted a systematic review and meta-analysis to estimate the prevalence of good adherence to statin monotherapy and to identify demographic and clinical factors associated with adherence among adults prescribed lipid-lowering therapy for the prevention of atherosclerotic cardiovascular disease. PubMed and Scopus were searched through May 2025 for randomized controlled trials, cohort, nested case control, and cross-sectional studies assessing statin adherence. Data were extracted on study design, participant characteristics, comorbidities, adherence measurement methods and duration, and statin type. Study quality was assessed using the Newcastle-Ottawa Scale, and risk of bias in randomized trials was evaluated with the Cochrane RoB 2 tool. Random-effect meta-analysis, along with subgroup and sensitivity analyses by follow-up duration, adherence thresholds, and study quality, was performed. Seventy-six studies including 5,898,141 participants with a median follow-up of 24 months were analysed. The pooled prevalence of good adherence ($\geq 80\%$ medication use) was 62.4% (95% CI: 58.3–66.5%), with lower adherence in primary prevention compared with secondary prevention settings. Lower adherence was associated with female sex, Black race, smoking, depression, and heart failure, whereas higher adherence was observed among older adults, those with myocardial infarction or hypertension, individuals with multiple comorbidities, and patients receiving polypharmacy. Primary non-adherence was not reported in any included study, and subgroup and sensitivity analyses showed consistent findings. Overall, statin adherence remains suboptimal and is strongly influenced by demographic and clinical factors, underscoring the need for targeted strategies to improve adherence, particularly among high-risk groups.

Lay summary

Statins are widely prescribed for the prevention and treatment of atherosclerotic cardiovascular disease (ASCVD), yet adherence to long-term therapy remains suboptimal. This systematic review and meta-analysis synthesizes evidence from 76 studies, including nearly 6 million individuals, to estimate adherence rates and identify the associated demographic and clinical determinants.

- Key finding 1: The pooled prevalence of good adherence (defined as $\geq 80\%$ medication consumption) was 62.4%, with lower adherence observed in primary prevention (57.5%) compared with secondary (established ASCVD) prevention setting (64.4%). Decreased adherence is noted among women, younger adults, individuals with depression, current smokers, persons of Black or Hispanic ethnicity, and heart failure patients.

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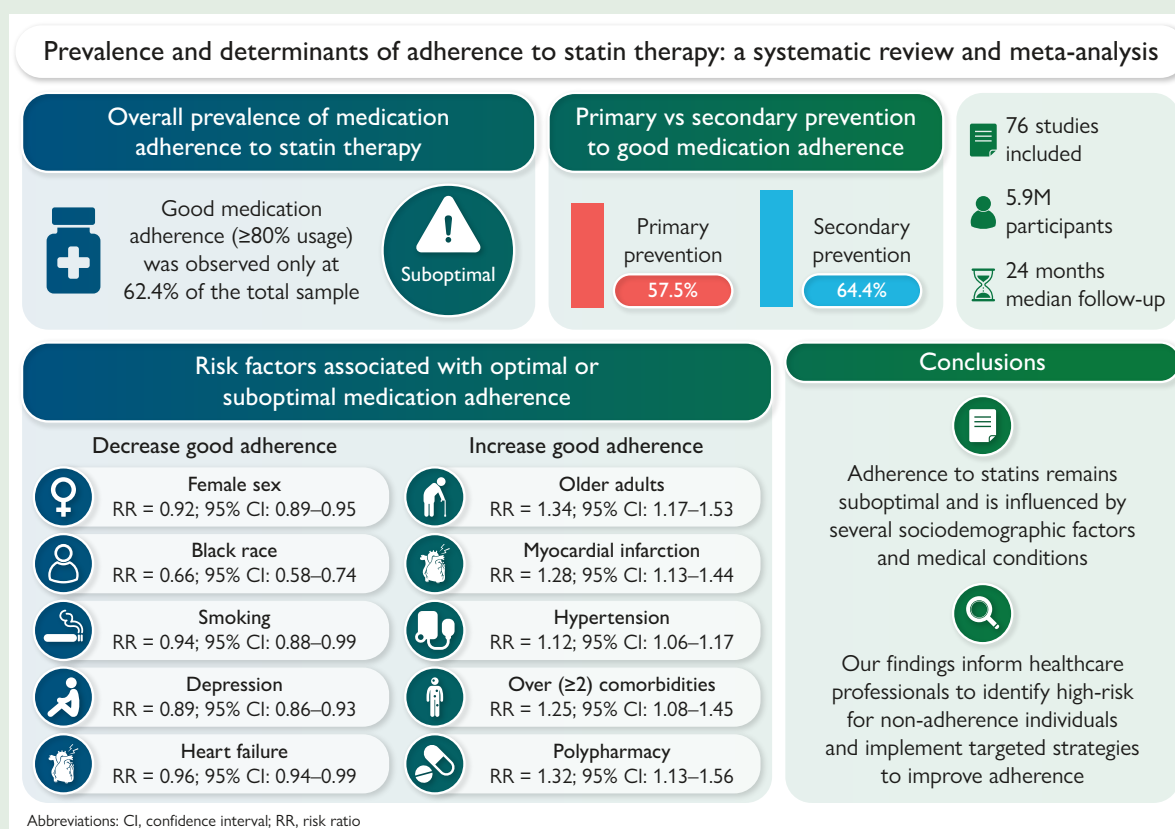
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- Key finding 2: Higher adherence is observed among older adults and those with a history of myocardial infarction, hypertension, or multiple comorbidities. Notably, polypharmacy (≥ 5 medications) is associated with greater adherence (a finding that contrasts with prior reports), which potentially reflects more intensive medical supervision or higher perceived cardiovascular risk in these populations and consorts with the higher adherence rates in those patients with established ASCVD.

These findings underscore the persistent challenge of ensuring sustained statin use, particularly among populations with higher ASCVD risk. Despite robust clinical guidelines and extensive evidence of benefit, nearly 4 in 10 patients fail to maintain adequate adherence levels. Given the association between poor adherence and increased cardiovascular morbidity and mortality, targeted interventions—especially for high-risk groups—are warranted. Understanding adherence patterns is essential for optimizing the real-world effectiveness of lipid-lowering therapies and improving long-term cardiovascular outcomes.

Graphical Abstract



Keywords

Statins • Medication adherence • Atherosclerosis • Cardiovascular diseases • Meta-analysis

Introduction

Atherosclerotic cardiovascular disease (ASCVD) remains the main cause of mortality worldwide.¹ Hypertension,² dyslipidemia,³ diabetes [type 2 (T2D), type 1 (T1D)],⁴ obesity,⁵ smoking,⁶ stress,⁷ physical inactivity,⁸ unhealthy diet,⁹ genetic, and environmental factors^{10,11} are related to ASCVD. Addressing and controlling these factors such as dyslipidemia with lipid-lowering treatment (LLT) based primarily on statins, is crucial to prevent ASCVD.^{12–14}

Adherence to therapy is another important but often overlooked issue. Reported rates of less than $<70\%$ ^{15–17} and even $<55\%$ for LLT specifically^{18–21} may impede the effectiveness of medications. Researchers frequently rely on medication adherence measures, which

quantify the proportion of medication taken as prescribed and accounts for persistence, that is, the duration a patient remains on therapy.²² Optimal medication adherence is defined when $\geq 80\%$ of the treatment is received based on pharmacy refill data, which showed the best prognostic discrimination.^{21,23–25} The common metrics for measuring medication adherence are the Proportion of Days Covered (PDC) and Medication Possession Ratio (MPR).²⁴

Poor adherence to LLT has substantial clinical implications. Patients with suboptimal adherence cannot achieve the target low-density lipoprotein cholesterol (LDL-C) levels and experience significantly higher rates of major acute cardiovascular events (MACE), as well as increased all-cause and cardiovascular (CV) mortality.^{26,27} Healthcare systems face higher burden of first and recurrent hospitalizations with the

subsequent economic cost.^{12,28,29} Therefore, improving adherence is crucial for patient outcomes and optimal use of healthcare resources.^{14,26,29–31}

Studies have identified several factors influencing medication-taking behaviour such as age,^{20,32–36} sex,^{20,32,36–39} race,^{40–42} socioeconomic factors,^{35,43,44} and comorbidities associated with atherosclerosis,^{34,39,45–49} but results are often inconsistent. Since the latest meta-analyses on the field,^{50–52} a considerable amount of evidence has been emerged.^{12,20,28,32–39,41–44,47,53–70} The aim of this systematic review and meta-analysis is to provide the most extensive and up to date synthesis of the evidence on medication adherence and a comprehensive analysis of risk factors affecting it, offering new insights and significantly enriching the existing literature.

Methods

This study is reported in accordance with the PRISMA guidelines⁷¹ (see [Supplementary material online, Table S6](#)) and has been registered at PROSPERO (CRD:420251000766).

Search strategy and eligibility criteria

We searched PubMed and Scopus (up to May 2025) for randomized and non-randomized controlled trials (RCT), cohort, case-control, and cross-sectional studies in humans that assessed the level of statin adherence (search algorithm in [Supplementary material online, Appendix](#)). The definition of adherence assessment is shown in [Supplementary material online, Tables S3 and S11](#).

Study quality assessment and risk of bias

We assessed study quality using the Newcastle-Ottawa Scale (NOS)⁷² for observational studies and for cross-sectional studies, which assesses Selection, Comparability, and Outcome domains. Selection ensured representative cohorts and validated adherence measurement. Comparability required controlling for demographic and CV factors. Outcome emphasized independent adherence assessment, adequate follow-up, and complete data. For randomized controlled trials, we used the Cochrane RoB 2 tool,⁷³ which assesses bias across six domains: randomization, deviations from intended interventions (assignment and adherence), missing data, outcome measurement, and result selection. Each domain is rated as low risk, some concerns, or high risk, informing the overall risk of bias (see [Supplementary material online, Table S4, S7, and S10](#)).

Statistical analysis

The two co-primary outcomes were (i) the prevalence (%) of good adherence to statin and (ii) medical, lifestyle, and socio-demographic factors affect it. Secondary outcome was the association of these factors with statin discontinuation.

Subgroup analyses explored variations in the risk of good adherence depending on the time window within which adherence was assessed (<1, 1, >1 year) and based on the criteria and overall NOS score (≤6 vs. >6). Sensitivity analyses were conducted by excluding studies using other than the standard adherence threshold (80%).

The DerSimonian–Laird random effects model was used when ≥3 independent studies were available. The degree of between-study variation on the estimates that can be attributed to heterogeneity was assessed using the I^2 metric.⁷⁴ The presence of small study effects, as an indication of publication bias, was investigated with the Egger’s regression asymmetry test ($P < 0.1$ indicates the presence of small study effects) and by visual inspection of the funnel plots (when ≥10 studies were available).⁷⁵ A P -value <0.05 was considered statistically significant. All analyses were performed using Stata software (version 16.1, Stata Corp LP).

Results

From the initially identified 10 185 articles, we finally included 76 eligible articles (see [Supplementary material online, Figure S1](#)). Details of the included studies are found in [Supplementary material online, Table S3 and S5](#). All studies had been published between 1998 and 2024. The range of sample size varied from 188 to 1 888 682 individuals, with a median of 944 435 participants, totalling 5 898 141 individuals. The age range varied from 18 to 86 years, with a median age of 52 years. Sixty-five studies focused only on medication adherence, two to medication adherence and discontinuation and nine only to discontinuation. Thirty-five studies were conducted in North America, 26 in Europe, 9 in Asia, 4 in Australia, and 2 in New Zealand. PDC was the most common metric of adherence (35 studies, 54%) compared with MPR (15 studies, 22%). Eleven studies assessed discontinuation rates of statin therapy, 10 studies assessed adherence in primary and secondary prevention population, 22 studies assessed adherence in primary prevention and 56 in secondary prevention.

Prevalence

The overall prevalence of good adherence to statins was 62.4% (95% CI 58.3%, 66.5%) from 67 studies including 5 124 410 individuals with an average follow-up of 24 months. Subgroup analyses assessing statin adherence stratified by countries/continent showed that the lowest prevalence of good adherence is evident in Asia and Canada (55% and 53%, respectively), followed by Europe (60%), and USA (65%), whereas the highest prevalence was noted in Oceania (75%) (see [Supplementary material online, Figure S60](#)). Adherence did not differ between countries with full/public healthcare coverage vs. private/limited coverage. Any differences based on income could not be assessed as only three countries belong to the non-high income group (see [Supplementary material online, Figures S61 and S62](#)). The prevalence of good adherence to statins among 1 134 199 individuals without established ASCVD (primary prevention) in 22 studies with an average follow-up of 22.9 months was estimated at 57.5% (95% CI 48.6%, 66.3%). In 56 studies comprising 4 044 254 individuals with established ASCVD (secondary prevention) of an average follow-up of 25 months, the prevalence of good adherence was 64.4% (95% CI 59.6%, 69.2%) (see [Supplementary material online, Figure S2](#)). The difference in adherence rates between primary and secondary prevention was not statistically significant (P for subgroup heterogeneity = 0.175).

Factors associated with statin adherence (Table 1, Figures 1–4)

- (1) Age groups: Older patients are more adherent compared with younger (≥65 vs. <65 years old RR = 1.34; 95% CI: 1.17–1.53; I^2 : 99%, five studies (see [Supplementary material online, Figure S3](#)).
- (2) Sex: Females are less adherent to statins compared with males (RR = 0.92; 95% CI: 0.89–0.95; I^2 : 98%, 53 studies) ([Figure 1](#)).
- (3) Race/ethnicity: Compared with White race, Black (RR = 0.66; 95% CI: 0.58–0.74; I^2 : 99%, 12 studies), Hispanic (RR = 0.71; 95% CI: 0.67–0.76; I^2 : 93%, six studies), Asian (RR = 0.83; 95% CI: 0.74–0.95; I^2 : 96%, six studies) or other race (RR = 0.81; 95% CI: 0.72–0.92; I^2 : 95%, nine studies) were associated with less adherence ([Figure 1](#)).
- (4) Existence of comorbidities: high vs. low (≥2 vs. <2) comorbidity score was not associated with increased adherence (RR = 1.11; 95% CI: 0.96–1.28; I^2 : 98%, five studies). Having one comorbidity

Table 1 Main results of the prevalence and factors associated to adherence

Prevalence				
Overall			62.38%	
Primary prevention ^a			57.46%	
Secondary prevention ^a			64.40%	
Factor	Comparison	RR (95% CI) ^b	I ²	Adherence
Age	≥65 vs. <65 years	1.34 (1.17–1.53)	99%	Increased
	<44 vs. 55–64 years	0.79 (0.67–0.93)	98%	Decreased
Sex	Females vs. Males	0.92 (0.89–0.95)	98%	Decreased
Race/Ethnicity	Black vs. White	0.66 (0.58–0.74)	99%	Decreased
	Latino vs. White	0.71 (0.67–0.76)	93%	Decreased
	Asian vs. White	0.83 (0.74–0.94)	95%	Decreased
	Other vs. White	0.81 (0.72–0.92)	96%	Decreased
Comorbidity score	High (≥2) vs. Low (<2)	1.11 (0.96–1.28)	98%	No effect
	2 vs. 0	1.25 (1.08–1.45)	97%	Increased
	1 vs. 0	1.13 (1.03–1.24)	92%	Increased
Myocardial infarction	Yes vs. No	1.28 (1.13–1.44)	97%	Increased
CVD/CAD/CHD	Yes vs. No	1.09 (0.97–1.23)	99%	No effect
Stroke	Yes vs. No	1.09 (0.94–1.25)	94%	No effect
Hypertension	Yes vs. no	1.12 (1.06–1.17)	94%	Increased
Polypharmacy	5–9 vs. <5 drugs	1.24 (1.16–1.32)	45%	Increased
	≥10 vs. <5 drugs	1.32 (1.13–1.56)	94%	Increased
Income	High vs. Low income	1.08 (0.98–1.19)	93%	No effect
Smoking	Smokers vs. Non-smokers	0.94 (0.88–0.99)	89%	Decreased
Depression	Yes vs. No	0.89 (0.86–0.93)	62%	Decreased
Diabetes	Yes vs. No	1.02 (0.97–1.06)	92%	No effect
Dementia	Yes vs. No	1.05 (0.94–1.17)	89%	No effect
Treatment intensity	High vs. Moderate/Low	0.94 (0.83–1.06)	96%	No effect
Chronic kidney disease	Yes vs. No	0.97 (0.89–1.06)	90%	No effect
Heart failure	Yes vs. No	0.96 (0.94–0.99)	0%	Decreased

CAD, coronary artery disease; CHD, coronary heart disease; CVD, cardiovascular disease.

^aPrimary vs. secondary prevention: without vs. with established cardio/cerebrovascular disease.^bBold face denotes statistically significant results.

was associated with increased adherence by 13% (RR = 1.13; 95% CI: 1.03–1.24; I^2 : 92%, three studies) and at least two comorbidities by 25% (RR = 1.25; 95% CI: 1.08–1.45; I^2 : 97%, three studies). Among comorbidities, myocardial infarction (RR = 1.28; 95% CI: 1.13–1.44; I^2 : 95.5%, 12 studies), hypertension (RR = 1.12; 95% CI: 1.06–1.17; I^2 : 97.5%, 21 studies), CVD/coronary artery disease (CAD)/coronary heart disease (CHD) (RR = 1.09 95% CI: 0.97–1.23; I^2 : 98.6%, 10 studies), and heart failure (RR = 0.96; 95% CI: 0.94–0.99; I^2 : 0%, five studies) were associated with increased adherence to statins, whereas stroke (RR = 1.09; 95% CI: 0.94–1.25; I^2 : 94%, 10 studies) did not (Figure 2, Supplementary material online, Figure S10).

- (5) Polypharmacy: Polypharmacy is positively associated with good adherence (5–9 vs. <5 different drugs RR = 1.24; 95% CI: 1.16–1.32; I^2 : 44.6%, and ≥10 vs. <5 different drugs RR = 1.32; 95% CI: 1.13–1.56; I^2 : 94%) (Figure 3).
- (6) Depression: Depression was negatively associated with adherence (RR = 0.89; 95% CI: 0.86–0.93; I^2 : 69%, 11 studies) (Figure 4).
- (7) Smoking: Current smoking habits were negatively associated with adherence (RR = 0.94; 95% CI: 0.88–0.99; I^2 : 89%, 16 studies) (Figure 4).

- (8) Diabetes (RR = 1.02; 95% CI: 0.97–1.06; I^2 : 92%, 22 studies), dementia (RR = 1.05; 95% CI: 0.94–1.17; I^2 : 89%, 10 studies), chronic kidney disease (CKD) (RR = 0.97; 95% CI: 0.89–1.06; I^2 : 90%, 12 studies), treatment intensity (RR = 0.94; 95% CI: 0.83–1.06; I^2 : 96%, 6 studies), and income (RR = 1.08; 95% CI: 0.98–1.19; I^2 : 93%, nine studies) did not seem to have any significant effect on adherence (see Supplementary material online, Figures S5 and S9).
- (9) Type of statins (atorvastatin vs. simvastatin and pravastatin vs. simvastatin) did not seem to have any significant effect on adherence—except when lovastatin was compared with simvastatin (RR: 1.14; 95% CI: 1.05–1.24; I^2 : 0%, three studies) (see Supplementary material online, Figure S4).

Sensitivity analyses assessed based on the time when adherence was assessed (<1 year, 1 year, >1 year) and excluded studies that used non-standard adherence thresholds (<80%), had NOS score ≤6, and did not meet the NOS criterion for sample representativeness showed no significant deviations from the main results, provided sufficient data were available (see Supplementary material online, Figures S11–S20 and S41–S56, Supplementary material online, Tables S1, S2 and S8). Visual inspection of funnel plots revealed no evidence of publication bias. Egger's test

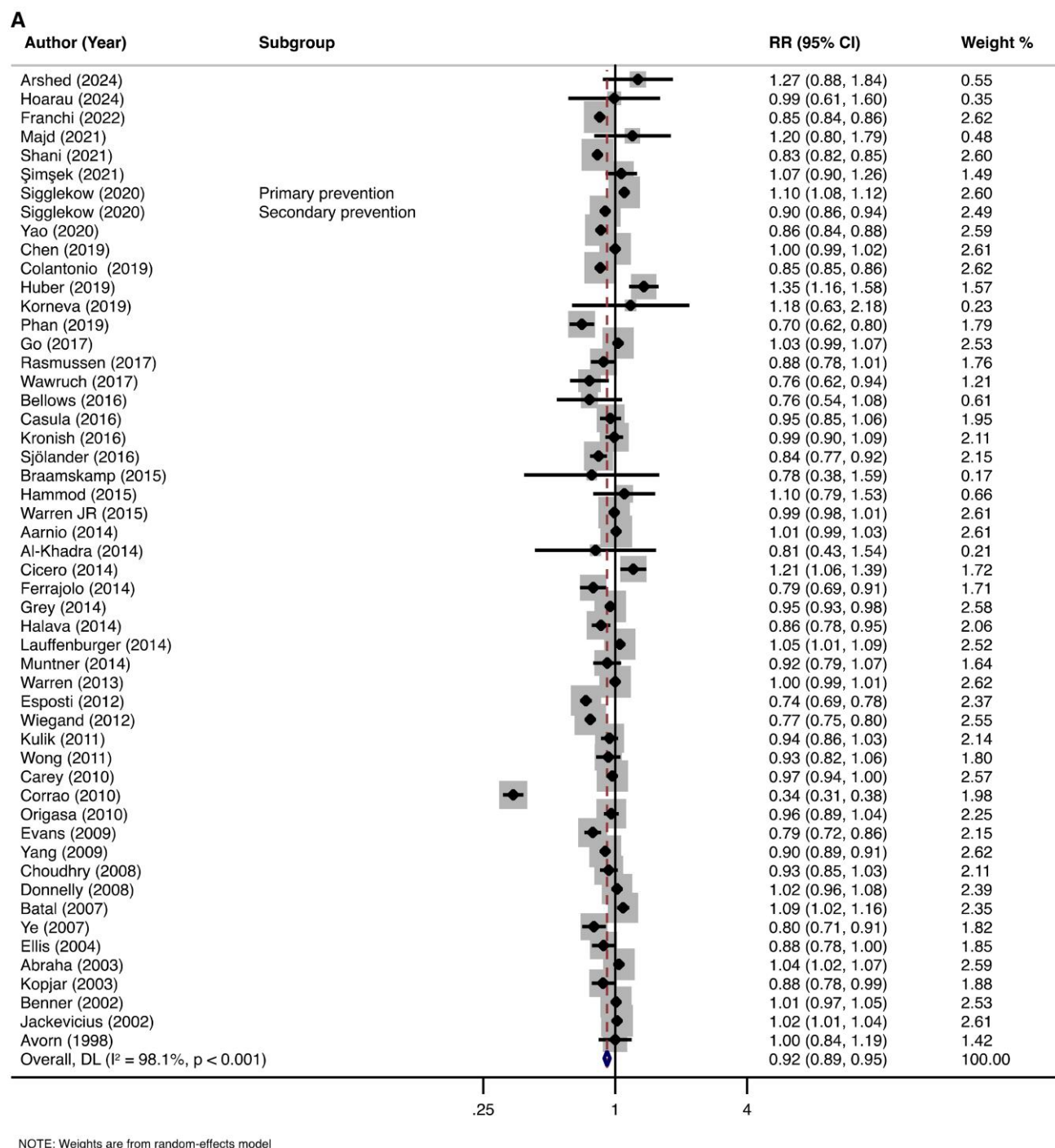


Figure 1 Demographic characteristics: (A) Sex, (B) Race and adherence. CI, confidence interval; RR, risk ratio.

(applied when ≥ 10 studies were available) showed no significant asymmetry (see [Supplementary material online, Figures S32–S40](#)).

Statin discontinuation

Eleven studies provided data on statin discontinuation. Hypertension was associated with 10% reduction in the likelihood of statin discontinuation, (RR = 0.90; 95% CI: 0.84–0.95; I^2 : 97%, five studies). Smoking, timing of adherence assessment, diabetes, race, sex, heart

failure, dementia, depression, CKD, and stroke were not associated with the risk of statin discontinuation (see [Supplementary material online, Figures S21–S31](#)).

Discussion

This comprehensive systematic review of the literature aimed to estimate the adherence to statin medication and identify its potential

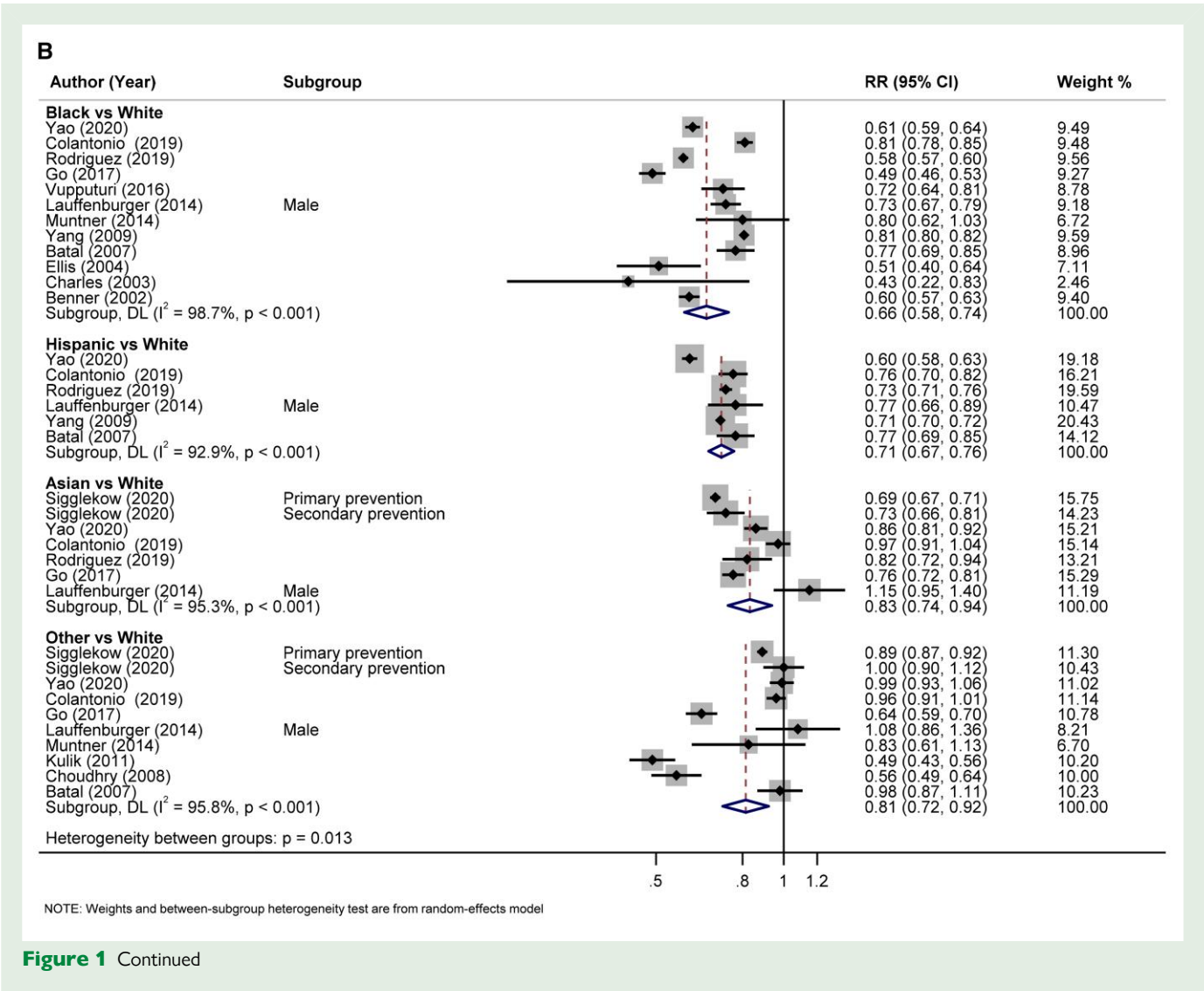


Figure 1 Continued

determinants. We observed that an average of only 62% of the patients were adherent to statins, which does not significantly differ between primary and secondary prevention setting, but differs by region/continent. While poor adherence to LLT is associated with adverse ASCVD events, which factors affect medication adherence have not been previously thoroughly evaluated. We identified a list of factors which may potentially affect adherence to statins, including age, sex, race/ethnicity, number and type of comorbidities, polypharmacy, smoking, depression, and heart failure.

Good adherence to a medication is defined when $\geq 80\%$ of the prescribed treatment is achieved.^{14,76} The common metrics of adherence to a medication are PDC and MPR or medication refill adherence (MRA).⁷⁰ The $\geq 80\%$ is clinically relevant as previous reports showed that when adherence $\geq 80\%$ is achieved, the risk for all-cause mortality decreases by 35%, for any CV event by 24%, for MI by 30% and for stroke by 32%.^{14,77}

Despite the accumulated evidence on the health benefits of statin use in both primary and secondary prevention of ACVD, the clear indications for their administration, their limited side effects, and the increasing use of social/educational initiatives and technological solutions (e-Health), statin adherence remains low.^{9,78} Our finding of only an

average of 62% of the participants in the primary achieving good adherence to statins corresponds to a population of only 3 out of 5 million individuals assessed.

A previous report based on significantly less (<15) studies including a total sample of 1.25 million participants similarly observed a suboptimal overall adherence of $<60\%$.²¹ Adherence rates differed by prevention setting with estimates of 57% for primary and 76% for secondary ASCVD prevention.⁷⁹ We similarly found that the adherence in primary prevention was lower (57%) compared with secondary prevention (64%), but now we show that for secondary prevention it is even lower compared with what was previously reported (64% vs. 76%, respectively). Among older adults (>65 years), 60% were reported to be adherent to statins at 1-year follow up, but only 61% of those that were initially adherent remained persistent at 4-years follow up.⁸⁰

Poor adherence to hypolipidemic therapy is associated with increased mortality,^{35,81,82} increased incidence of stroke,^{19,83} and poor prognosis of any atherosclerotic disease.^{15,17,84} Specifically, good vs. poor ($<80\%$) adherence was associated with a lower ASCVD risk by 15% and all-cause mortality by 45%. The estimated absolute risk difference for any ASCVD associated with poor medication

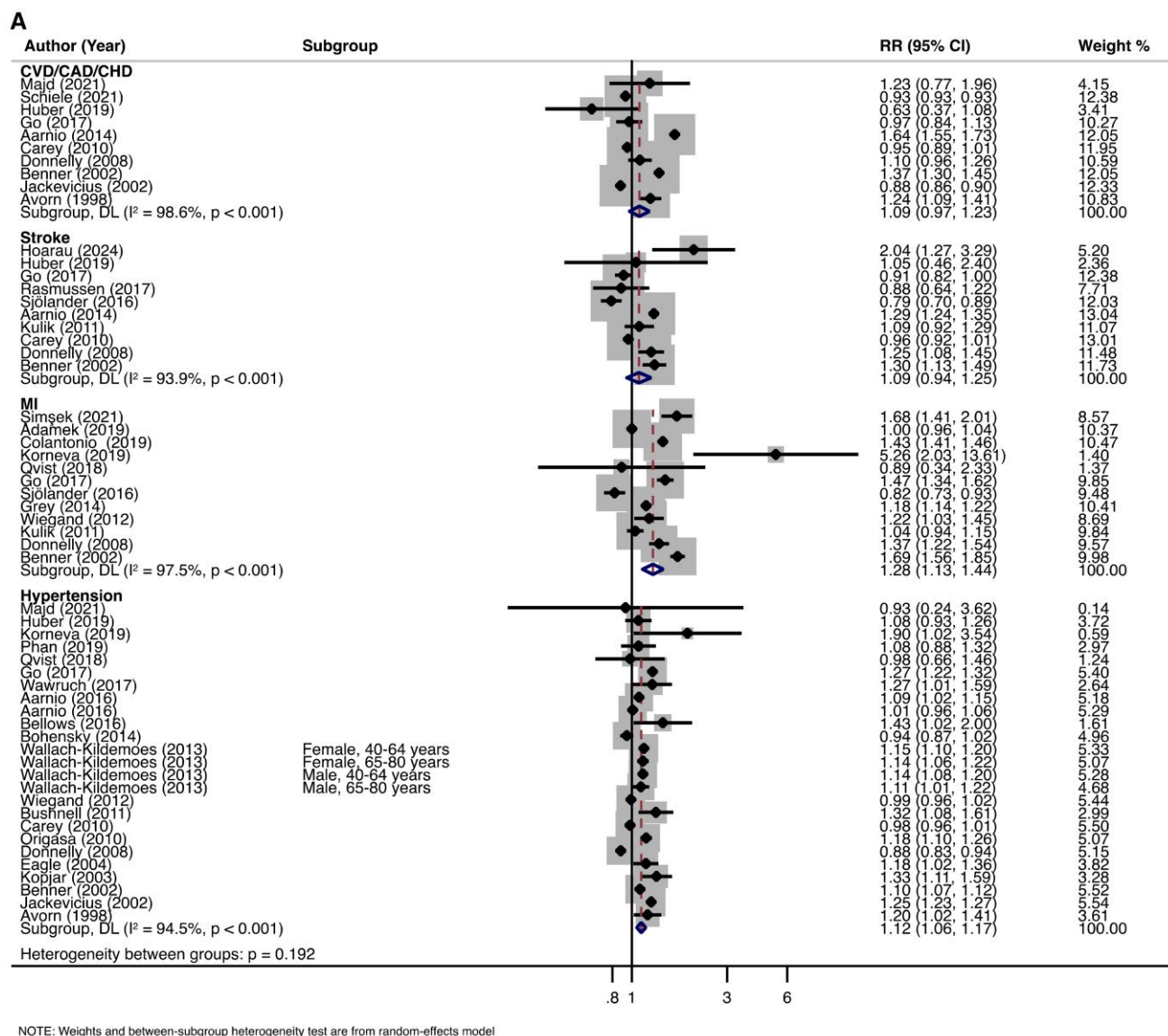


Figure 2 (A) Comorbidity type, (B) Comorbidity score and adherence. CI, confidence interval; RR, risk ratio.

adherence were nine cases for statins per 100 000 individuals per year.²¹

For those patients with established CAD, good adherence to statin reduced mortality by 15% and any ASCVD event by 10%. The estimated absolute differences per 1 million individuals per year for mortality associated with 20% improvement in adherence were 175 cases for statin, which outweighed the benefit observed with other medications used in secondary prevention setting.⁸⁵ Similarly, the risk of ischaemic stroke for the categories with the highest compared with the lowest adherence to statins was 0.83, resulting in 7% lower stroke risk with an improvement by 20% in statin adherence.⁸⁶ When adherence to evidence-based medication regimens [not only statins but also b-blockers, angiotensin-converting-enzyme inhibitors/angiotensin 2 receptor blockers (ACEi/ARB), antiplatelets] was assessed in patients with stable CAD, good adherence was related to a lower risk of all-cause mortality by 44%, ASCVD mortality by 34%, and CV hospitalization/MI by 39%.⁸⁷

Identified socio-demographic factors associated with the likelihood of statin adherence and discontinuation included age, sex, and race. We observed that females seem to have a lower likelihood of being adherent to statins and a higher risk to discontinue statins compared with males. This might be related to the fact that males traditionally have a higher risk of developing ASCVD compared with females^{50,88} and our results indicate that history of MI or hypertension is associated with higher adherence to LLT. Younger age seems to be associated with poor adherence, which is partly attributed to the increased CV risk and the development of ASCVD during ageing.⁵⁰ Regarding race, our findings are also in agreement with the existing evidence which shows that Caucasians are the most adherent compared with other races. Income has been previously reported to affect medication adherence, with individuals at higher income (>70 000 USD) levels being more likely to be adherent to their medication compared with lower (<30 000 USD).^{33,41,47,50,55} In our study, we did not find any association between income and adherence.

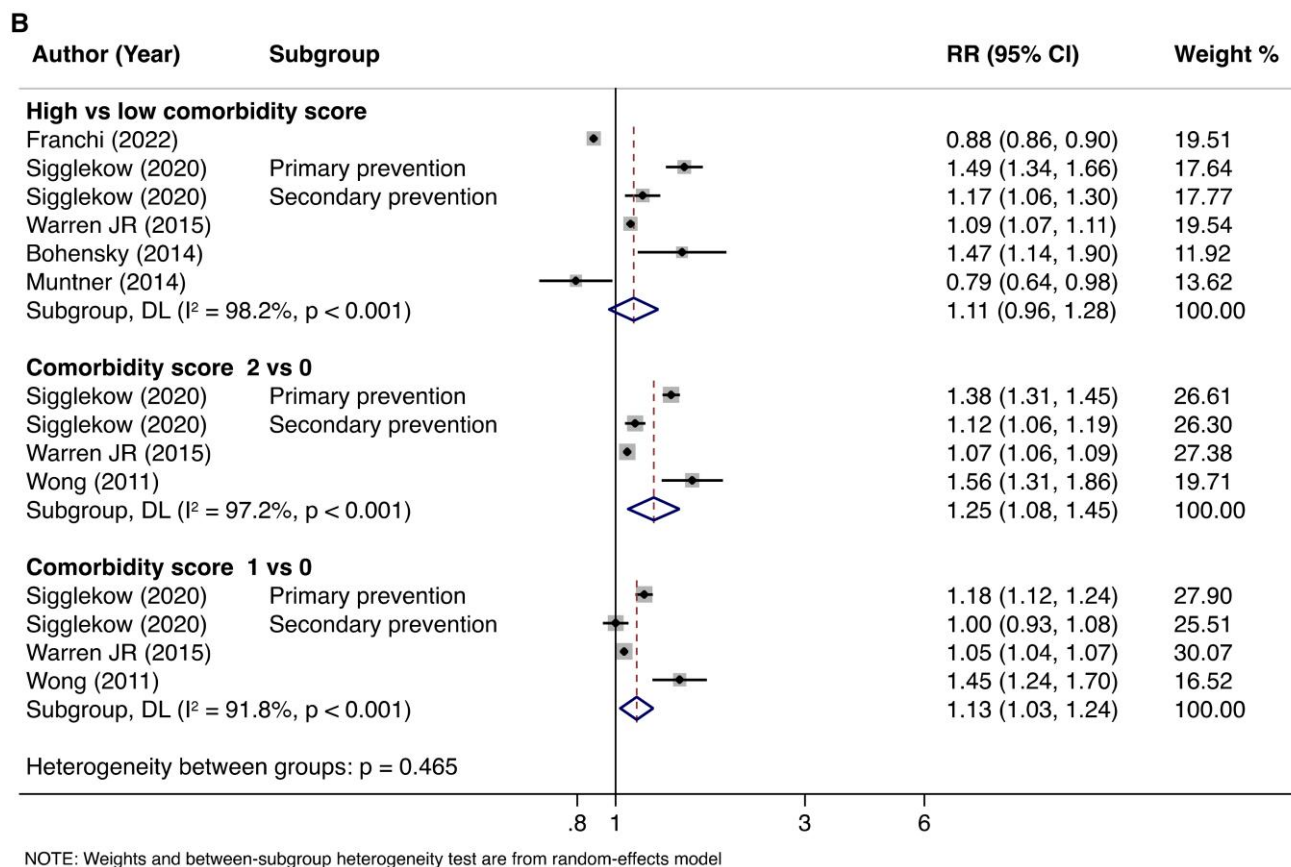


Figure 2 Continued

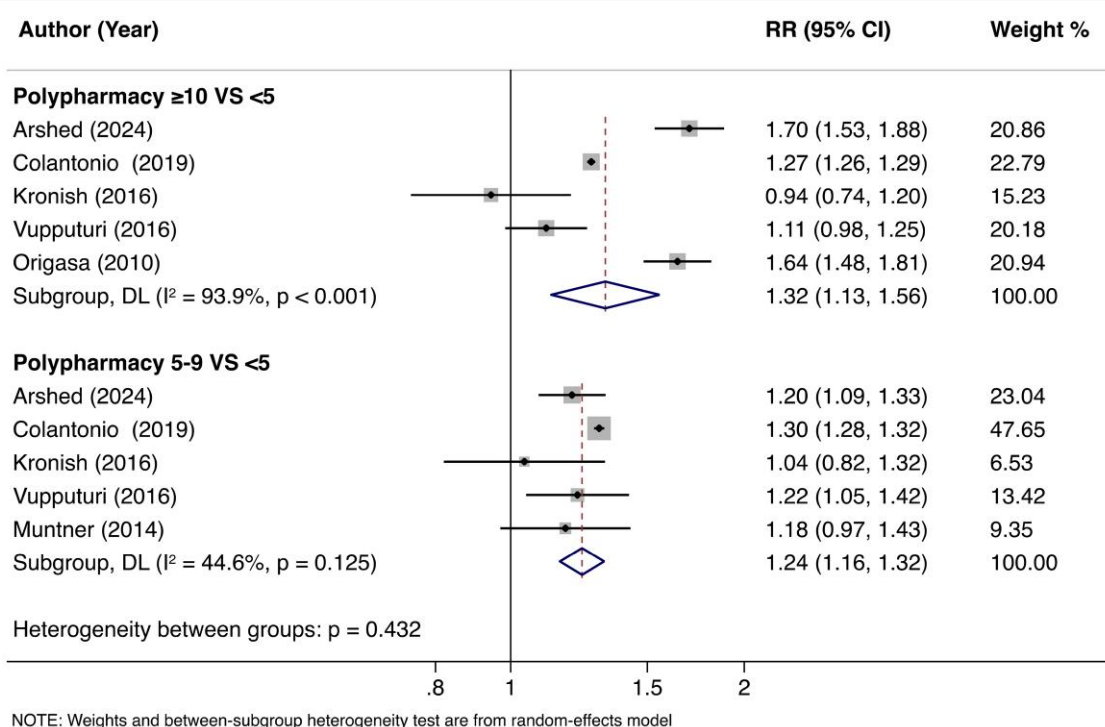


Figure 3 Polypharmacy and adherence. CI, confidence interval; RR, risk ratio.

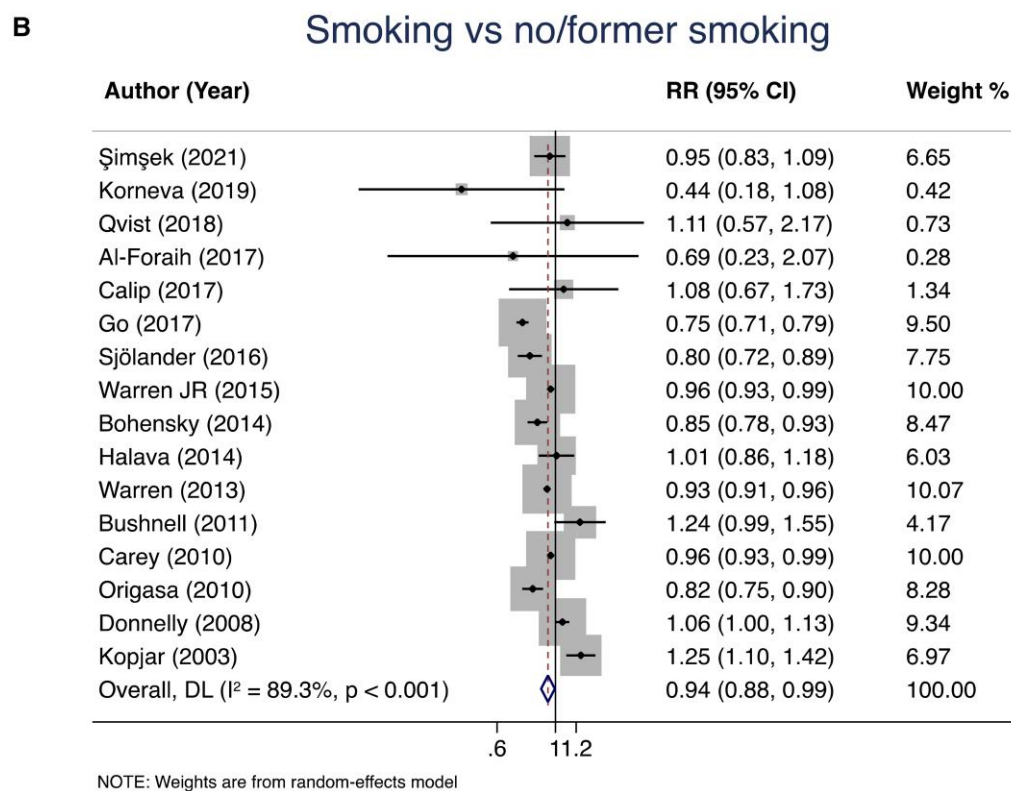
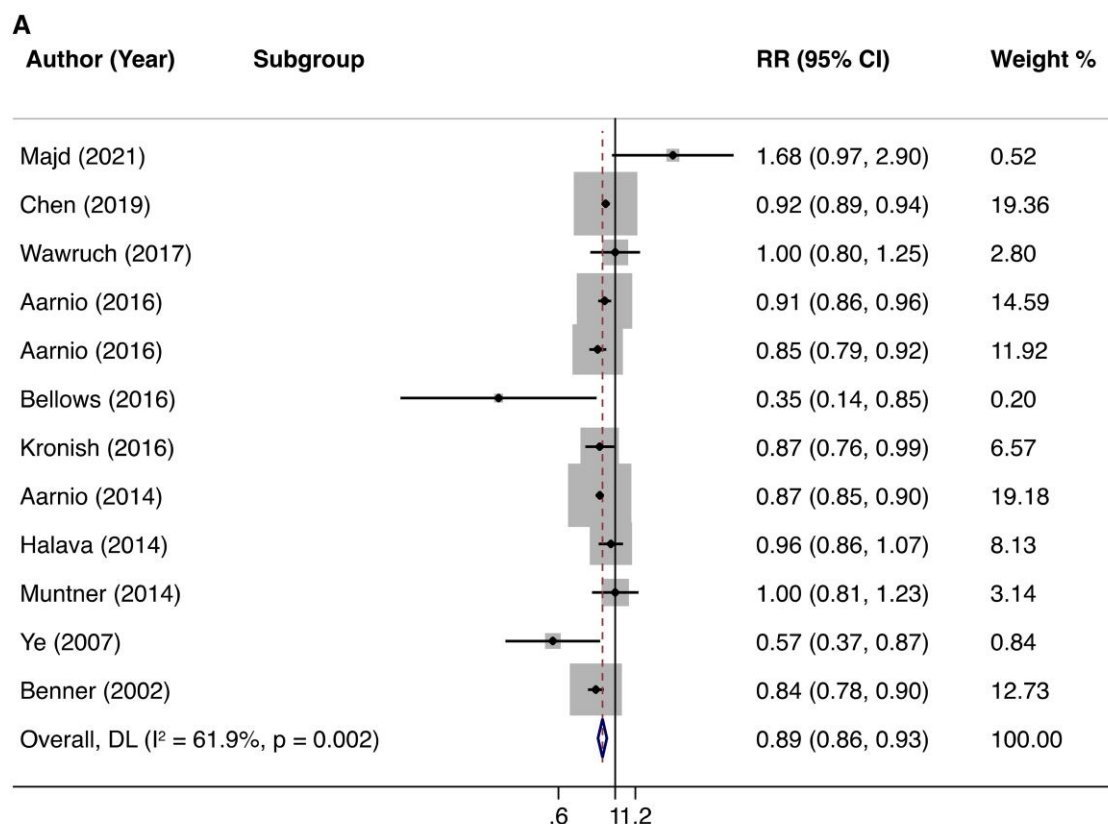


Figure 4 (A) Depression, (B) Smoking and adherence. CI, confidence interval; RR, risk ratio.

Having at least one comorbidity increases the likelihood of good statin adherence,^{32,33,57,70} especially when the comorbidities are associated with ASCVD.^{39,45,46,20,34,37,42,47–50,55} Although polypharmacy is often considered a barrier to adherence, our systematic review suggests that polypharmacy is associated with increased medication adherence. Possible explanations are the greater disease burden (evident in established ASCVD), the closer clinical monitoring, and the heightened patient awareness of the importance of statin therapy when comorbidities are present.^{52,89} Our results and those of previously published studies indicate that polypharmacy should be viewed in the context of the types of comorbidities.^{12,20,41} Of note, we emphasize that we assessed polypharmacy and statin adherence, and our results cannot be generalized to polypharmacy and its relationship with any other (non-statin) medication. In contrast to previous publications, which showed that people who receive high-intensity statins seem to have higher chances of being adherent^{35,43,44} (related to the fact that more potent statins are used in high CV risk individuals), we failed to replicate this finding. This might be explained by the fact that not only statin type but also statin dose is equally important when addressing ASCVD risk. Finally, mental health has been reported to affect adherence.^{12,28,37,90,91} In our study, we found that depression, a condition not previously reported to affect adherence, is another limiting factor for an individual to be adherent to LLT.

Since the latest meta-analyses in the field,^{50–52} an additional 32 new studies accounting for 3.9 million participants update our knowledge on statin adherence and explain some of the above discrepancies. Among older (>65 years) adults, the factors previously associated with decreased statin adherence (such as age >75 years, non-White race, female sex, current smoker, new user, lower number of concurrent CV medications, primary prevention setting, depression, and not having renal disease) do not substantially differ from our results.^{52,80} Beyond statin adherence, we additionally provide data on statin discontinuation, showing for the first time that females are more prone to complete discontinuation of statins compared with males, contrary to the existence of hypertension, which decreases the chance of statin discontinuation.

LLT is crucial for the prevention and management of ASCVD, which is the leading cause of death worldwide.^{14,81,92} Prevention and management of ASCVD using LLTs can only be effective if individuals adhere to their prescribed medications. To actively increase medication adherence, knowledge of its barriers and facilitators is important.^{93–97} By knowing which factors are commonly associated with poor medication adherence, healthcare professionals could identify those subjects who are prone to be less adherent and adopt strategies to increase persistence to medication. Poor adherence, a significant public health challenge impacting ASCVD prevention, management, and healthcare costs, underscores that the success of LLT relies not only on the development of more advanced therapies but also on ensuring consistent adherence to proven treatments.⁷⁷

Limitations of this meta-analysis, which are largely consistent with those commonly encountered in large-scale meta-analyses, include the design of the included studies [very few RCTs (3), many prospective and retrospective studies (68)], the high heterogeneity as assessed with I^2 , the inclusion of studies which did not have data solely on statin use but included also other drugs (anti-hypertensives, beta-blockers, antiplatelets, etc.) and the diversity of the tools used to assess adherence. Nevertheless, funnel plots indicated no evidence of publication bias, Egger's tests imply that results are reliable and not influenced by small-study effects or selective publication of only significant results, sensitivity analyses failed to show a differentiation of the results compared with the main analyses, and studies included in the meta-analysis had high NOS score (median score 7, 4 studies with score

<6 were excluded). 'Primary' non-adherence—defined as the failure to initiate prescribed therapy—was not reported in any of the included studies. Furthermore, we were unable to identify enough studies to evaluate the impact of dosage on statin adherence. Finally, most studies reported adherence as a dichotomous outcome, which may oversimplify the complexity and variability of adherence behaviour.

Conclusions

Overall adherence to statins remains suboptimal and is adversely influenced by several socio-demographic and medical factors such as female sex, young age, non-White race, smoking, depression, and heart failure. Myocardial infarction, hypertension, ≥ 2 comorbidities, and polypharmacy are associated with higher adherence. Our findings could support and inform healthcare professionals to identify high-risk for non-adherence individuals and implement targeted strategies to improve adherence.

Supplementary material

Supplementary material is available at *European Journal of Preventive Cardiology*.

Author contributions

AB and ECR contributed to the conceptualization of the study. AB, GM, and ECR were responsible for the development of the methodology. Formal analysis was performed by AB, GM, and CAC. GM contributed to software development. The investigation was carried out by AB, GM, CAC, and ECR, while data curation was handled by AB, GM, and KC. Validation was undertaken by AB, GM, KC, CAC, EL, KKT, and ECR. Resources were provided by GM, EEN, and KKT. Visualization was conducted by AB and ECR. The original draft of the manuscript was prepared by AB and ECR, and all authors (AB, GM, EEN, KC, CAC, EL, KKT, MP, and ECR) contributed to the review and editing of the manuscript. Project administration was led by AB and ECR, and ECR also supervised the overall conduct of the study.

All authors have read and approved the final manuscript and agree to be accountable for all aspects of the work.

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Data availability

The data underlying this article are available in the article and in its online [Supplementary material](#).

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